THE MOVING OBSERVER

Can a changing viewpoint improve exoplanet access?

Biswajit Jana | Finding Earth 2.0

Independent research and simulation project | 24 October 2026 | Discussion brief v7.1

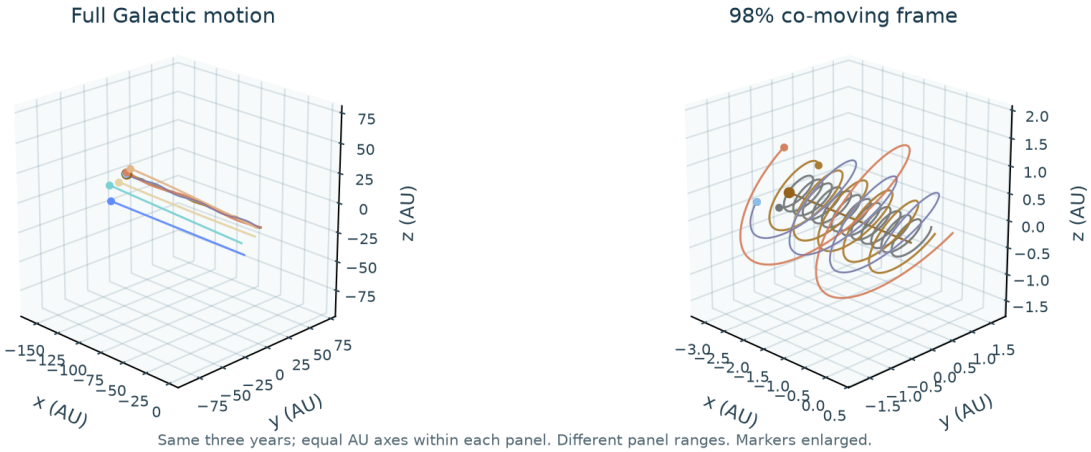
The question

The Solar System moves through the Milky Way, while neighbouring stars follow their own orbits. Could changing **relative** positions make particular exoplanets easier to observe? I built this project to separate three effects: Galactic travel, motion relative to a target, and improvements in an instrument.

What the present calculation says

Selected Gaia sample: 1,387 sources	Accessibility sum
2026 geometry; baseline thresholds	29.63207
2126 geometry; same thresholds	29.64118
2026 geometry; improved thresholds	102.72300

Century motion adds about 0.03% to the baseline score. The assumed instrument change has a much larger effect. These are single-visit accessibility scores for an injected planet distribution - **not counts of discovered or habitable planets**. Some threshold choices reverse the sign of the motion effect.



Same three years; equal AU axes within each panel. Different panel ranges. Markers enlarged.

Why the frame matters

In the current model the Sun moves about 4,917 AU in a century. Nearby stars move too. A released probe inherits its velocity and follows gravity; it does not remain at a fixed Galactic coordinate. A fixed 2026 marker is a hypothetical reference. Winding planetary paths depend on the chosen frame.

A request for scientific guidance

I am an independent researcher without a formal supervisor. My aim is to make the question testable and invite correction, not to claim a new physical principle.

Three questions for an expert

1. Is the comparison useful? Which prior or follow-up work best addresses changing observer-side accessibility at fixed instrument capability?

2. Which targets survive a fuller model? Do any gains remain after binary dynamics, astrometric systematics, planet populations, exposure times and observing schedules are included?

3. Would a second observer help? What feasible probe trajectory and measurable observable would improve on Earth-orbit and long-term astrometry?

How to explore the simulation

Milky Way: play the 250 Myr illustration; compare radius, height and a labelled smooth-density scenario. Synthetic example stars are off by default.
Sun & all planets: choose the winding-path tour, then compare full Galactic and 98% co-moving frames. Use the separate Solar-wobble and Earth-Moon close-ups.
Space & Time: compare a stationary clock with ideal circular-orbit clocks around a 3D Earth. The 400 km and 20,200 km presets give about -24.7 and +38.5 microseconds/day relative to a stationary surface clock.

Limits and the next test

The Galaxy artwork is not a measured star-count map. The existing illustrative potential is not a fit to the latest rotation curve. Smooth density does not predict spiral-arm crossings. The Moon is a circular illustration; the relativistic model assumes an isolated, nonrotating spherical body. The next research step is a small, carefully chosen real-target sample with a realistic observing budget and uncertainty propagation.

Selected foundations

Kaltenegger & Faherty (2021): changing Earth transit-zone membership.

Bailer-Jones (2022): stellar encounters from Gaia DR3.

Stark et al. (2024): robust HWO exoplanet yield margins.

Poggio et al. (2025): Galactic vertical corrugation.

JPL Horizons; Gaia DR3; NIST clock tests. Full provenance is in the report.

Simulation, full report and reproducible source
biswajit1999.github.io/moving-observer/
github.com/Biswajit1999/moving-observer